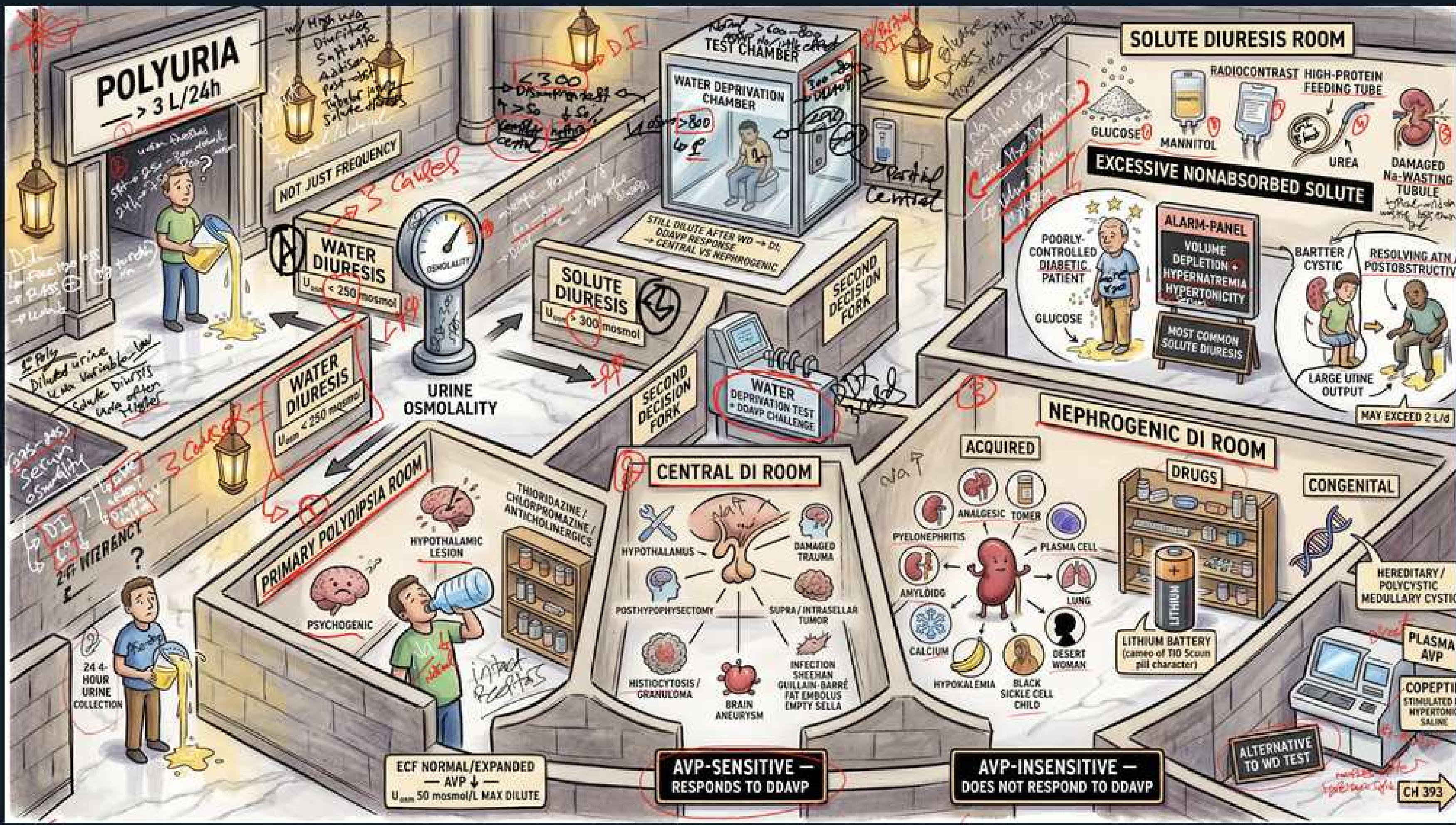
Polyuria — Water vs Solute Diuresis & DI Subtypes



1. Polyuria > 3 L/24h

WHAT YOU SEE

Sign 'POLYURIA >3 L/24h', man at urinal.

WHAT IT ENCODES

Polyuria is urine output >3 L/day, distinct from frequency (small volumes) and nocturia. Confirm with a 24-hour collection before workup. Splits into water diuresis vs solute (osmotic) diuresis by urine osmolality.

BOARD TRAP

Urinary frequency is NOT polyuria — small frequent voids point to bladder/prostate, not a renal concentrating defect; measure the 24h volume.

2. Urine osmolality splits the diagnosis

WHAT YOU SEE

Gauge 'URINE OSMOLALITY' between Water and Solute diuresis.

WHAT IT ENCODES

Urine osm <300 mOsm/kg = water diuresis (DI or primary polydipsia); >300 with high solute excretion = solute diuresis. This single number is the first fork in the polyuria algorithm.

BOARD TRAP

A dilute urine (<300) excludes osmotic diuresis — don't chase glucose/mannitol when the urine is hypotonic; that's a water-handling problem.

3. Water diuresis

WHAT YOU SEE

Sign 'WATER DIURESIS', low urine osmolality.

WHAT IT ENCODES

Water diuresis = large volumes of dilute (hypotonic) urine from DI (can't concentrate) or primary polydipsia (suppressed AVP). >24h urine collection confirms; water-deprivation/copeptin testing separates the causes.

BOARD TRAP

Water diuresis means dilute urine — high urine osmolality rules it OUT and points instead to a solute (osmotic) load.

4. Solute / osmotic diuresis

WHAT YOU SEE

Sign 'SOLUTE DIURESIS', water deprivation chamber.

WHAT IT ENCODES

Solute diuresis = >600–900 mOsm/day excreted, dragging water out: hyperglycemia (glucose), mannitol, high-protein tube feeds (urea), post-obstructive, radiocontrast. Urine osm stays >300.

BOARD TRAP

Osmotic diuresis from glucosuria/urea won't correct with DDAVP — fix the solute load, don't give vasopressin.

5. Excessive nonabsorbed solute — glucose, mannitol, urea

WHAT YOU SEE

Panel 'EXCESSIVE NONABSORBED SOLUTE': glucose, mannitol, urea, high-protein feeding tube.

WHAT IT ENCODES

Classic osmotic loads: uncontrolled diabetes (glucose), mannitol infusion, high-protein enteral feeding and radiocontrast (urea). Each delivers nonreabsorbed solute to the tubule, obligating water loss.

BOARD TRAP

High-protein tube feeds cause a urea osmotic diuresis with rising BUN — mistaken for DI if you don't check the solute load.

6. Damaged/resisting tubule — Bartter, cystic, post-obstructive ATN

WHAT YOU SEE

Damaged tubule cartoon: Bartter, cystic, post-obstructive ATN.

WHAT IT ENCODES

Tubular injury impairs reabsorption: Bartter/Gitelman, medullary cystic disease, recovering ATN, and post-obstructive diuresis all cause inappropriate salt/water wasting. Large urine output may exceed 3–6 L/day.

BOARD TRAP

Post-obstructive and recovering-ATN diuresis is self-limited tubular wasting — over-replacing fluids perpetuates it; match output, don't chase it.

7. Water deprivation chamber

WHAT YOU SEE

Water deprivation test chamber.

WHAT IT ENCODES

Water deprivation withholds fluid to see if endogenous AVP concentrates the urine; failure to concentrate then a response to DDAVP indicates central DI, while no response indicates nephrogenic DI.

BOARD TRAP

Water deprivation is being supplanted by copeptin testing — and is dangerous in true DI (risk of hypernatremia); monitor weight and Na closely.

8. Central DI room — AVP-sensitive

WHAT YOU SEE

Brain/pituitary cartoons, 'CENTRAL DI ROOM', AVP-SENSITIVE responds to DDAVP.

WHAT IT ENCODES

Central DI = deficient AVP from hypothalamic/posterior pituitary damage (trauma, surgery, tumor, infiltration, infection). Urine osm rises >50% after DDAVP. Treat with desmopressin.

BOARD TRAP

Central DI responds robustly to DDAVP; nephrogenic does not — the DDAVP response, not the deprivation alone, separates them.

9. Nephrogenic DI room — AVP-insensitive

WHAT YOU SEE

Sign 'NEPHROGENIC DI ROOM', AVP-INSENSITIVE, does not respond to DDAVP.

WHAT IT ENCODES

Nephrogenic DI = kidney resistance to AVP. Acquired causes (lithium, hypercalcemia, hypokalemia, pyelonephritis) and rare congenital V2-receptor/AQP2 defects. Minimal urine concentration after DDAVP.

BOARD TRAP

Failure to respond to DDAVP = nephrogenic, not central — giving more desmopressin won't help; remove the offending drug/electrolyte.

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10. Lithium — most common acquired nephrogenic DI

WHAT YOU SEE

Lithium battery labeled most common acquired cause.

WHAT IT ENCODES

Chronic lithium is the most common acquired cause of nephrogenic DI; it enters principal cells via ENaC and downregulates AQP2. Amiloride blocks lithium entry; stop lithium if possible.

BOARD TRAP

In lithium NDI, amiloride (not more thiazide alone) blocks ENaC-mediated lithium uptake — and never restrict water in a polyuric DI patient.

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11. Primary polydipsia

WHAT YOU SEE

Sign 'PRIMARY POLYDIPSIA', patient drinking compulsively.

WHAT IT ENCODES

Primary (psychogenic/dipsogenic) polydipsia = excess water intake suppresses AVP, causing dilute polyuria with LOW-normal serum sodium. Urine concentrates appropriately with deprivation; AVP/copeptin are appropriately low.

BOARD TRAP

Primary polydipsia gives LOW serum Na, whereas DI tends toward HIGH-normal/elevated Na — the baseline sodium distinguishes them before any test.

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12. Hypercalcemia / hypokalemia → nephrogenic

WHAT YOU SEE

Calcium and potassium cues in nephrogenic room.

WHAT IT ENCODES

Hypercalcemia and hypokalemia impair medullary concentrating ability and AQP2 trafficking, producing reversible acquired nephrogenic DI. Correct the electrolyte to restore concentrating capacity.

BOARD TRAP

Polyuria from hypercalcemia mimics primary DI but reverses once calcium normalizes — check Ca/K before labeling it idiopathic.

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13. 24h urine collection confirms

WHAT YOU SEE

Bottom-left '24h urine collection', man voiding.

WHAT IT ENCODES

Always document true polyuria (>3 L/day) with a timed collection before extensive testing. Separates genuine polyuria from frequency/nocturia and anchors the water-vs-solute fork.

BOARD TRAP

Skipping the 24h volume risks working up 'DI' in a patient who merely has frequency — confirm the volume first.

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14. Alarm panel — diabetic/volume depletion

WHAT YOU SEE

Red 'ALARM PANEL' poorly-controlled diabetic, volume depletion.

WHAT IT ENCODES

Uncontrolled diabetes is the prototypical solute diuresis: glucosuria obligates water and electrolyte loss, causing volume depletion and possibly large urine output. Glycemic control resolves it.

BOARD TRAP

Glucosuric osmotic diuresis won't respond to DDAVP and shows urine osm >300 — treat the hyperglycemia, not a presumed water-handling defect.
